

Stellar PSFs Fake Hosts in Little Red Dots

Mismatched spectra between source and field stars systematically inflate inferred host sizes, luminosities, and masses.

arXiv:2608.30459 — Source-conditioned polychromatic PSFs for compact high-redshift nuclei

First author / year / source

arXiv:2608.30459 doi:10.48550/arXiv.2608.30459



Background

Compact high-redshift sources are modelled by subtracting a point-spread function built from nearby field stars. This assumes the target's spectrum matches the stellar reference across every filter.

No one had quantified how spectral mismatch between source and PSF stars biases broad-band host measurements.

What this paper is selling

The authors derive the physical and mathematical origin of this light-misassignment error and demonstrate a correction. Applied to little red dots, it revises host sizes, luminosities, and masses.

Key results to remember

- A pure point source generates spurious wavelength-dependent Sérsic residuals in every filter.
- The published 11.8% F444W host component survives no tested spectral or positional variation.
- Source-conditioned polychromatic PSFs remove the bias for other compact distant sources.

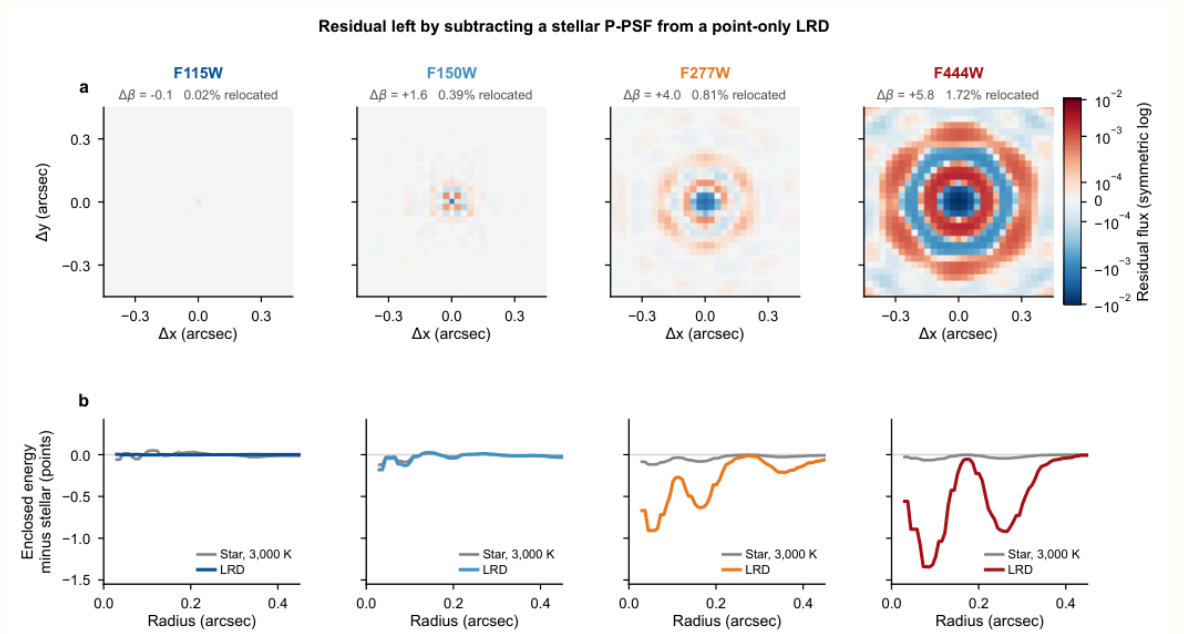


Figure 1 | A point-only LRD produces the wavelength-dependent residual assigned to a Sérsic component.

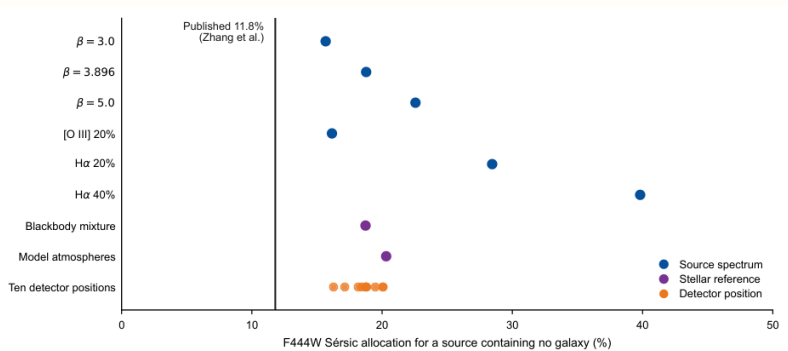


Figure 2 | The F444W allocation for a source containing no galaxy survives every tested change of input.